A geometric computation of cohomotopy groups in co-degree one

Michael Jung

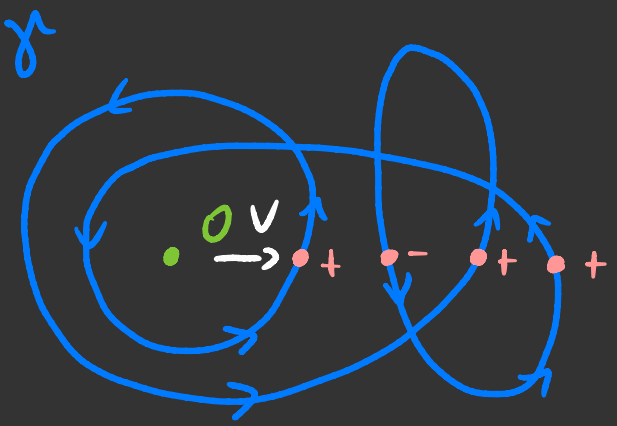
joint work with
Thomas Rot

Vrije Universiteit
Amsterdam

July 14, 2023

1 Baby example

Q: Curves $\gamma : S^1 \to \mathbb{R}^2 - \{0\}$ up to homotopy?



γ has winding number

$$w(\gamma) = 2.$$

Define $f : S^1 \to S^1$ via

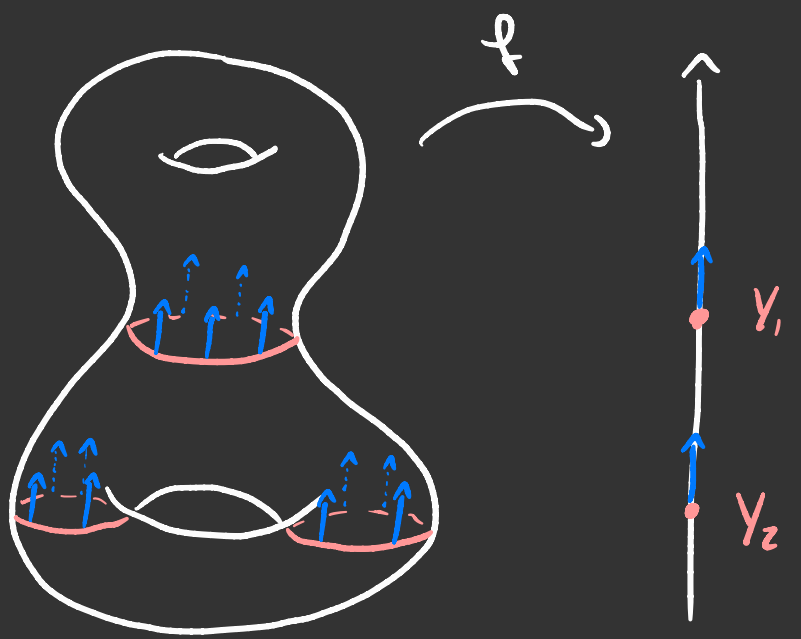
$$f(t) := \frac{\gamma(t)}{\|\gamma(t)\|}.$$

Then: $$w(\gamma) = \sum_{t \in f^{-1}(v)} \operatorname{sgn} df|_t = 1 - 1 + 1 + 1 = 2.$$

$=: \deg(f; v)$

mapping degree

1 Regular values



$f: X \to M$ smooth

$\Rightarrow f^{-1}(y) \subset X$ submfd.
if y is reg. value

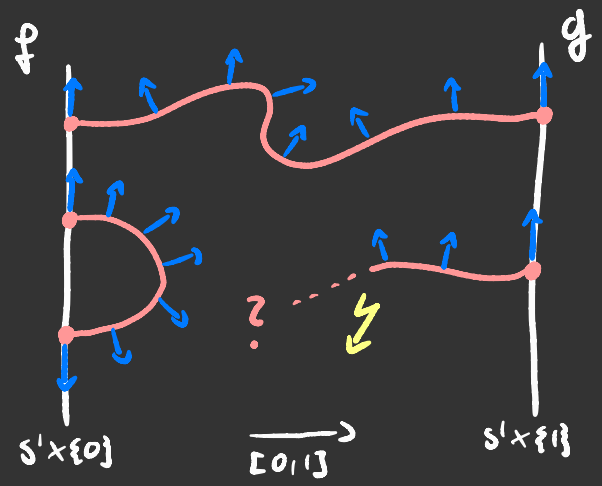
AND $\nu_{f^{-1}(y)} \cong f^*(T_y M)$

$\Rightarrow f^{-1}(y)$ has triv. norm. bundle in X !

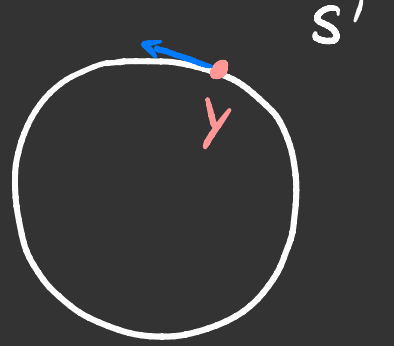
Fact: Reg. values are "generic".

Back to the baby example

Q: Is mapping degree a homotopy invariant?





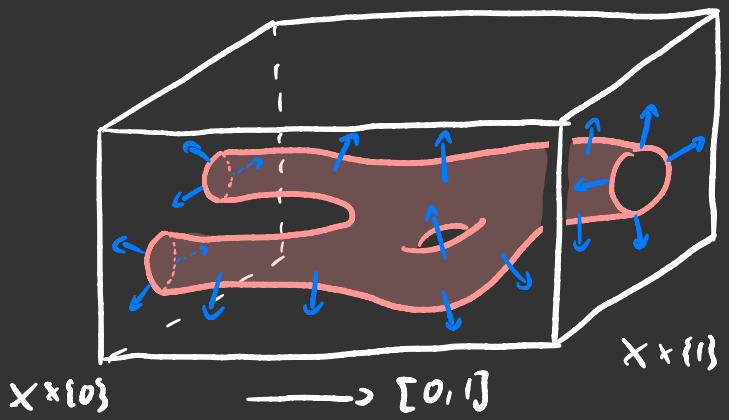


Assume $\deg(f; y) \neq \deg(g; y)$.

Pontryagin-Thom construction

Q: Homotopy invariant for $f: X \to S^n$?

Remember: $f^{-1}(y) \subset X$ submfd. $\rightsquigarrow$ cobordisms?



Define:

$$\mathbb{F}_k(X) := \frac{\{k\text{-dim closed } M \subset X \times \mathbb{R}^\infty \text{ framing of } \nu_M\}}{\text{normally framed cobord.}}$$

Note: $\mathbb{F}_k(X)$ is group if $\dim X \gg k$:

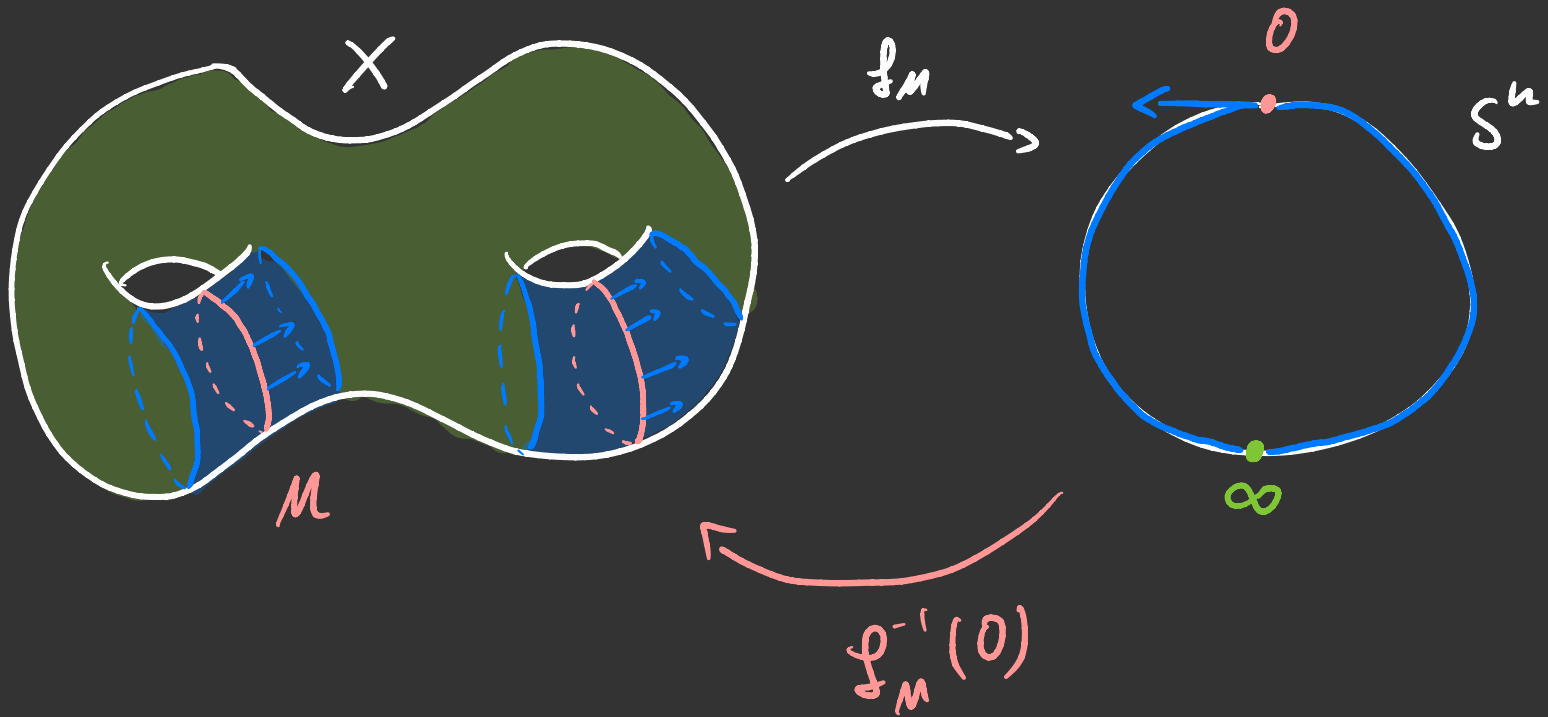
i) $[M_1, \varphi_1] + [M_2, \varphi_2] = [M_1 \sqcup M_2, \varphi_1 \sqcup \varphi_2]$

ii) $-[M, \varphi] = [M, -\varphi]$ (reverse orientation)

Pontryagin-Thom construction



A: $[X^{n+k}, S^n] \cong \Pi F_k(X)$



What is $[X^{n+1}, S^n]$ (aka $\mathbb{F}_1(X)$)?

- Enumerated by Steenrod '47
- Group ext. by Taylor '09 but purely algebraically
- Geometric proof for $n=3$ and X^4 orientable mfd. by Kirby, Melvin, Teichner '12
- Geometric proof for $n \geq 3$ and X^{n+1} spin mfd. by Konstantis '20

Q: Geometric proof for $n \geq 3$ and X^{n+1} (non-orientable) manifold?

1 Twisted homology

$$\mathbb{L}_1(X; \mathbb{Z}_w) := \frac{\{\text{Links } L \subset X \text{ w/ orientation of } \nu_L\}}{\text{normally oriented bordisms}}$$

"Algebraically":

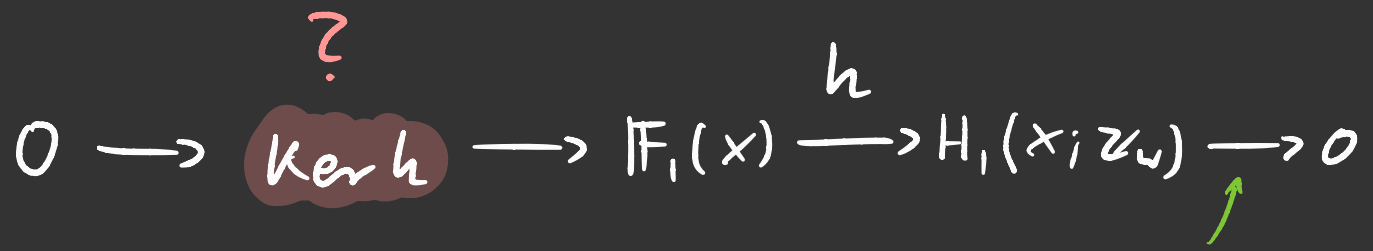
$$\mathbb{L}_1(X; \mathbb{Z}_w) \cong H_1(X; \mathbb{Z}_w) = \text{first homology w/ "twisted" coefficients}$$

(Thom '54, Atiyah '60)

↳ obvious map $h : \mathbb{F}_1(X) \longrightarrow H_1(X; \mathbb{Z}_w)$

forgets framing, remembers orientation

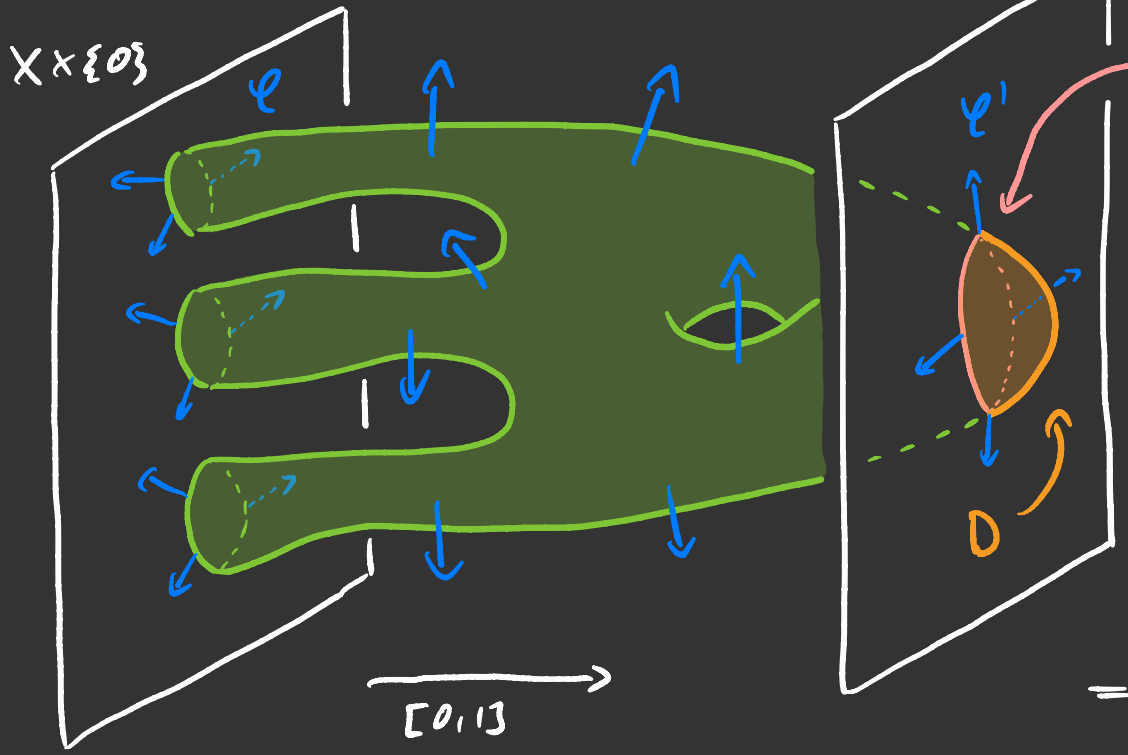
1 The short exact sequence



Also: what is the extension?

Determine $\ker h$

Suppose $[L, \varphi] \in \ker h$.



U contractible

$\rightsquigarrow$ induces (unique) framing φ' over U s.t. $[L, \varphi] = [U, \varphi']$.

Now:

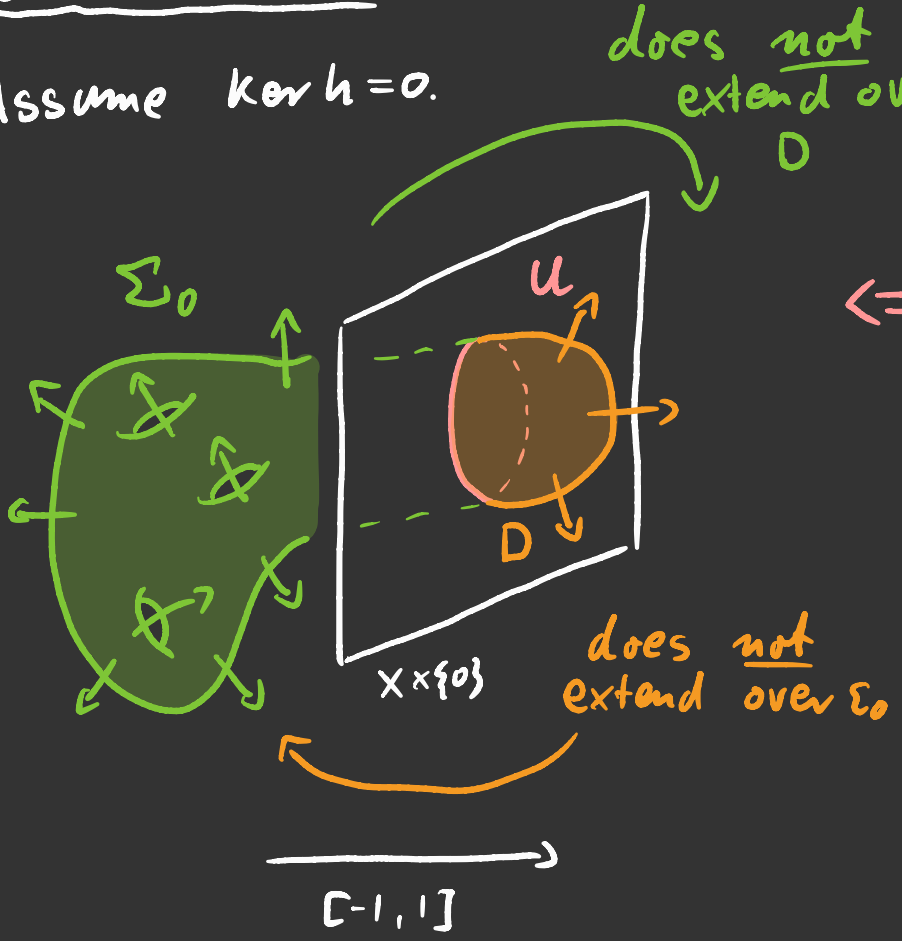
$\pi_1(SO(n)) = \mathbb{Z}_2$

$\Rightarrow$ U has <u>two</u> possible framings (up to homotopy)

$\Rightarrow$ $\ker h$ is <u>at most</u> $\mathbb{Z}_2$!

Determine $\ker h$

Assume $\ker h = 0$.



$\Sigma = \Sigma_0 \cup D$ surface w/ ν_ξ or. but not triv.

$\Longleftrightarrow$

say X is type I iff $\exists\, \Sigma \subset X$ s.t. ν_ξ or. but not triv.

First result

Theorem: X is type I iff $h : \mathbb{F}_1(X) \longrightarrow H_1(X; \mathbb{Z}\omega)$ is an isomorphism.

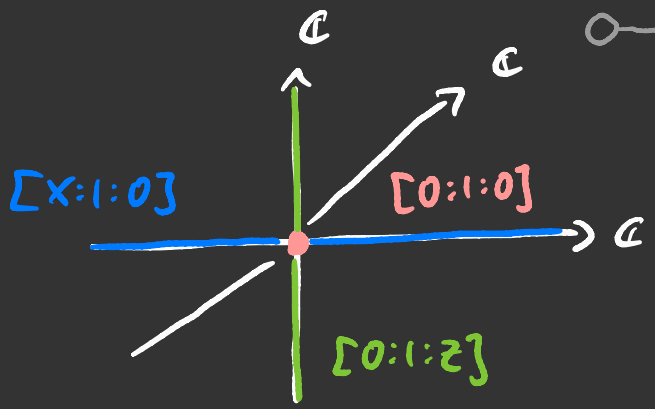
And define:

X is type II iff $\forall \Sigma \subset X$ surface

ν_Σ or. $\Rightarrow$ ν_Σ triv.

1 Examples

1) $\mathbb{C}P^1 \subset \mathbb{C}P^2$



$\rightsquigarrow$ $\mathbb{C}P^1$ has odd self-intersection

$\Rightarrow$ $\mathbb{C}P^2$ is of typeI

$\rightsquigarrow$ $IF_1(\mathbb{C}P^2) \cong H_1(\mathbb{C}P^2; \mathbb{Z}_w) \cong H_1(\mathbb{C}P^2; \mathbb{Z}) = 0$

2) $\mathbb{R}P^2 \subset \mathbb{R}P^{4k}$ $\xrightarrow{\text{char. classes}}$ $\nu_{\mathbb{R}P^2}$ or. but non-triv.

$\rightsquigarrow$ $IF_1(\mathbb{R}P^{4k}) \cong H_1(\mathbb{R}P^{4k}; \mathbb{Z}_w) \cong 0$ all circles C with orientable ν_C are contractible

Result for type II

Theorem: If X is type II, then $\mathbb{F}_1(X)$ fits into

$$0 \to \mathbb{Z}_2 \longrightarrow \mathbb{F}_1(X) \to H_1(X;\mathbb{Z}_w) \to 0.$$

The extension is uniquely determined by $w_1^2(X) + w_2(X)$.

$w_1^2(X) + w_2(X)$: Pin^--obstruction class!

In particular:

$\mathbb{F}_1(X) \cong H_1(X;\mathbb{Z}_w) \oplus \mathbb{Z}_2$ iff X is Pin^-.

Proof idea



Fact: Extension of

$$0 \to \mathbb{Z}_2 \to \Gamma_1(X) \to H_1(X;\mathbb{Z}_w) \to 0$$

is (uniquely) determined by hom.

$$\varepsilon_X : \underbrace{\mathrm{Tor}_2(H_1(X;\mathbb{Z}_w))}_{=:\{[c]\in H_1(X;\mathbb{Z}_w) \mid 2[c]=0\}} \to \mathbb{Z}_2.$$

← "measures triviality of ext. in 2-torsion part"

Link to geom. of X:

1) $[c] \in \mathrm{Tor}_2(H_1(X;\mathbb{Z}_w))$ circle.
2) Endow c with normal framing ℓ.
3) Then $2[c,\ell]=0$ iff $\varepsilon_X([c])=0$.
4) Build up extension via generators.

Proof idea

$[C]_{tw} \in \mathrm{Tor}_2(H_1(X;\mathbb{Z}_w))$ generator

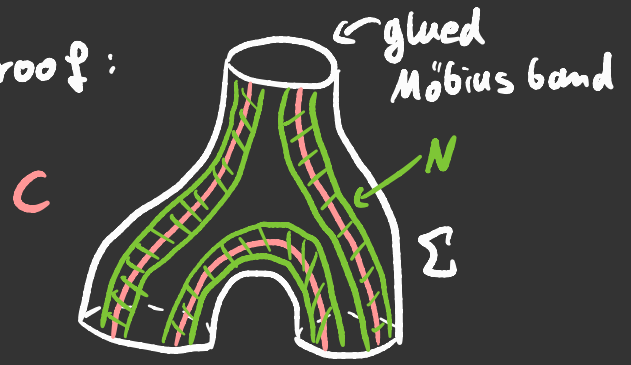
$\exists\ \beta : H_2(X;\mathbb{Z}_2) \longrightarrow \mathrm{Tor}_2(H_1(X;\mathbb{Z}_w))$ surjective s.t.

$\beta([\Sigma]_2) = [C]_{tw}$ and C represents $w_1(\gamma_\Sigma)$

Prop: $2[C,\varphi] = 0$ iff $w_2(\gamma_\Sigma) = 0$

$2[C,\varphi] = [\partial N, \psi]$

Proof:



Proof idea

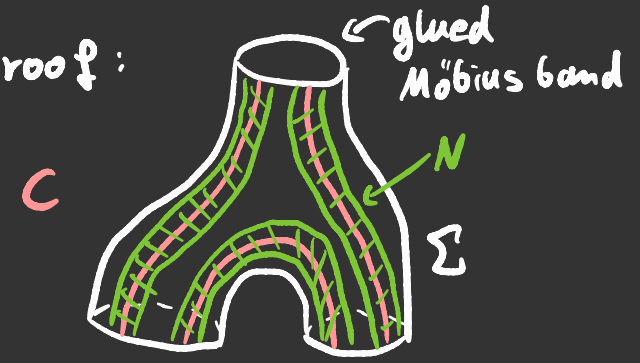
$[C]_{tw} \in Tor_2(H_1(X; \mathbb{Z}_w))$ generator

$\exists\ \beta : H_2(X; \mathbb{Z}_2) \longrightarrow Tor_2(H_1(X; \mathbb{Z}_w))$ surjective s.t.

$\beta([\Sigma]_2) = [C]_{tw}$ and C represents $w_1(\gamma_\Sigma)$

Prop: $2[C, \varphi] = 0$ iff $w_2(\gamma_\Sigma) = 0$

Proof:



$2[C, \varphi] = [\partial N, \psi]$

Now: $[\partial N, \psi] = 0$ iff ψ extends over $\Sigma - N$

Obstruction theory:

ψ extends over $\Sigma - N$ iff $w_2(\gamma_\Sigma) = 0$

□

Example

Consider $X = \mathbb{RP}^{4+1}$ and $\mathbb{RP}^2 \subset \mathbb{RP}^{4+1}$

$[\mathbb{RP}^2]$ generates $H_2(X; \mathbb{Z}_2)$ and $\mathrm{Tor}_2(H_1(X;\mathbb{Z}_2)) = H_1(X;\mathbb{Z}_2)$.

$(4+1) \bmod 4$	0	1	2	3
$w_1(\nu_{\mathbb{RP}^2})$	0	1	0	1
$w_2(\nu_{\mathbb{RP}^2})$	1	1	0	0
type	I	II, not Pin^-	II, Pin^-	II, Pin^-
$\pi^4(\mathbb{RP}^{4+1})$	0	$\mathbb{Z}_4$	$\mathbb{Z}_2$	$\mathbb{Z}_2 \oplus \mathbb{Z}_2$

1 Pin⁻ manifolds

X has Pin⁻-structure, then

$$0 \longrightarrow \underset{\cong \mathbb{Z}_2}{\Omega_1^{\mathrm{Pin}^-}} \longrightarrow \mathbb{F}_1(X) \longrightarrow H_1(X;\mathbb{Z}_w) \longrightarrow 0$$

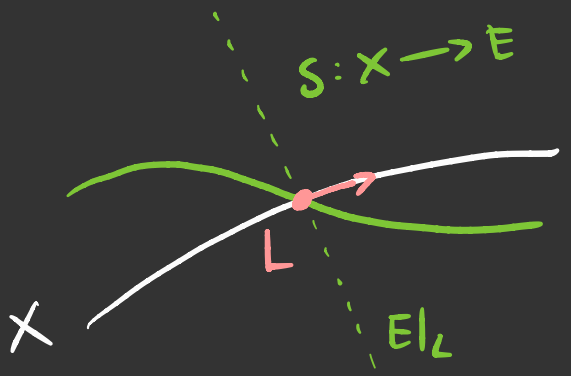
induced splitting map

Then:

$$\begin{array}{ccc} \mathrm{Pin}^-(X) \times H^1(X;\mathbb{Z}_2) & \xrightarrow{\text{act}} & \mathrm{Pin}^-(X) \\ \downarrow & & \downarrow \\ \mathrm{Spin}(X) \times \frac{H^1(X;\mathbb{Z}_2)}{\langle w_1(X) \rangle} & \xrightarrow{\text{act}} & \mathrm{Spin}(X) \end{array}$$

Applications to vector bundles

$\mathbb{R}^n \hookrightarrow E \to X^{n+1}$ w/ orientation & spin structure



$L := s^{-1}(0_E)$

spin structure

$\rightsquigarrow \quad \nu_L \cong E|_L \overset{\varphi}{\cong} L \times \mathbb{R}^n$

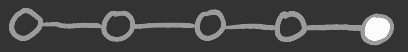
$\rightsquigarrow \quad [L, \varphi] \in \mathbb{F}_1(X)$

$\hookrightarrow \quad h([L,\varphi]) =$ PD of Euler class $e(E)$

Theorem: E admits a non-vanishing section if and only if $[L, \varphi] = 0$.

Idea: Use null-bordism to "push" s away from zero (similar to whitney trick).

Applications to vector bundles



Corollary 1: If X is of type I, then E admits a non-vanishing section iff $e(E) = 0$.

Corollary 2: If X is Pin$^-$, then E admits a non-vanishing section iff $e(E) = 0$ and $\mathfrak{Sq}^2([L, e]) = 0$.

$\mathfrak{Sq}$ splitting map
$\mathfrak{Sq}: \mathbb{F}_1(X) \to \mathbb{Z}_2$

THANK YOU FOR YOUR ATTENTION!

